

Navigating exponential polyhedra without solving LPs

The most promising path combines three mutually reinforcing ideas: dual-based cluster bounding from parametric LP sensitivity, ejection chains that traverse infeasible intermediates by design, and spectral decomposition of the constraint graph to identify natural macro groupings. These methods—drawn from operations research, scheduling theory, and algebraic graph theory—can be layered into a multi-fidelity evaluation cascade that screens candidates at sub-millisecond cost before committing to any LP solve. The structural analog closest to this problem is Graphs of Convex Sets (GCS) from robotics motion planning, which formalizes exactly the “union of convex regions” optimization and provides tight convex relaxations. A realistic 6-week implementation plan would prioritize the evaluation cascade and ejection chain infrastructure, with spectral decomposition providing the backbone for intelligent neighborhood selection.

The macro placement problem’s constraint structure—the line graph $L(K_n)$ of the complete graph—has a known three-level eigenvalue spectrum that directly reveals the natural decomposition hierarchy. Combined with recent breakthroughs in constraint-matrix sensitivity analysis and practical expander decomposition algorithms, this creates an actionable framework for characterizing entire clusters of polyhedra cheaply.

Bounding LP families without solving them: the Miftari approach and beyond

The single most directly applicable result is the 2024–2026 work by Miftari, Derval et al. (arXiv:2410.14443, v3 February 2026) on sensitivity analysis for linear modifications of the LP constraint matrix. Their framework studies $f(\lambda) = \min c^T x$ subject to $(A + \lambda D)x \leq b$, where D encodes constraint-matrix perturbations— [Optimization-online](#) precisely what happens when switching a separation direction from L to R changes specific rows of the constraint matrix. Their **linear Lagrangian bound** uses dual variables p^* from a solved LP to cheaply bound the optimal value as constraints change, requiring only matrix-vector products rather than full LP solves. [Optimization-online](#) Their companion paper (arXiv:2501.04151, January 2025) extends this with Neumann series expansions: if $\|(\lambda + \delta)E_B\|_\infty < 1$, the basis inverse can be approximated and the objective deviation bounded analytically. [arXiv](#)

The practical recipe is: solve one “center” LP per cluster, extract the optimal dual vector π^* ,

then for each nearby LP differing in k separation directions, check whether π^* remains dual feasible. If so, $\pi^{*\top} b_{\text{new}}$ is a valid lower bound—computed in microseconds. When the bound exceeds the best known solution, the entire cluster is pruned without any additional LP solve. Empirical testing should measure how tightness degrades as k increases from 1 to 10 constraint switches. **The expected failure mode is basis instability:** when switching a direction affects a tight (active) constraint, the bound becomes vacuous. Degeneracy in the LP compounds this issue. [UBalt](#)

Logic-Based Benders Decomposition (LBBD), comprehensively treated in Hooker’s 2024 Springer monograph, provides the principled framework for accumulating pruning information across evaluations. The master problem selects L/R/A/B assignments; each LP subproblem generates Benders cuts in the assignment-variable space. [ResearchGate](#) After solving several subproblems, the accumulated cuts yield valid lower bounds for *any* assignment, including unvisited ones. The June 2025 work on Disjunctive Benders Decomposition (arXiv:2506.03561) enhances this by generating valid inequalities for the convex hull of the Benders reformulation, tested successfully on large-scale facility location instances. [arXiv](#)

Surrogate duality adds another layer. Müller et al. (arXiv:1912.00356, published Mathematical Programming 2022) revisit generalized K -surrogate relaxation for mixed-integer programs: aggregating m constraints into K weighted constraints yields bounds strictly tighter than $K=1$ Lagrangian relaxation, computed via a Benders-based algorithm.

[Springer](#) [Springer](#) For cluster evaluation, one aggregates the “switched” constraints using surrogate multipliers, replacing k disjunctive constraints with a single tractable LP.

Minimal viable experiment: Solve one LP. Record dual variables. For 1,000 nearby assignments (differing in 1–10 pairs), compute the Miftari linear Lagrangian bound and dual-feasibility-based bound. Compare both to actual LP solutions. This characterizes the “LP landscape smoothness” and directly reveals whether cluster pruning is viable. Estimated setup time: 3–5 days.

Ejection chains solve the infeasibility problem by design

The core barrier—that group flips of 200+ pairs create circular constraint chains causing LP infeasibility—is precisely the problem that ejection chain methods were designed to handle. Ejection chains (Glover, Rego; Annals of Operations Research 2010) create a reference structure that is *deliberately infeasible*, allowing a sequence of moves to propagate before feasibility is restored. At each step of the chain, a trial solution is available by “ejecting” the current infeasibility through a specific repair.

For the disjunctive assignment, the ejection chain operates as follows. Starting from a “pivot

pair" whose direction change has the largest potential impact, the change propagates through dependent pairs via BFS/DFS on the constraint graph. Each dependent pair's direction is flipped to maintain local consistency. At depths 1, 5, 10, 20, trial solutions are constructed by repairing unassigned pairs using fast constraint propagation, and the LP is evaluated only for the most promising trial. The key innovation over random multi-pair flips is **sequential construction**: rather than flipping 200 pairs simultaneously and hoping for feasibility, the chain builds the new assignment incrementally, maintaining a feasibility-restoring "escape route" at every step.

Adaptive Large Neighborhood Search (ALNS) provides the complementary destroy-and-repair framework. [ScienceDirect](#) The BALANCE framework (Phan et al., 2023) uses Thompson sampling for adaptive operator selection, achieving 50% improvement in operator choice quality. For macro placement, the "destroy" operator removes 5–20% of pair assignments (in a spatially coherent region identified by spectral clustering), and the "repair" operator uses constraint propagation plus a fast heuristic before LP warm-starting. The key practical insight across all methods in this category: **navigate through infeasible intermediate states using cheap heuristic evaluation, reserving LP solves for feasibility-restored trial solutions.** [ScienceDirect](#)

Two-Dimensional Parallel Tempering (arXiv:2506.14781, 2025) extends replica exchange along a second penalty-strength axis in addition to temperature, with feasibility-based swaps transferring configurations from soft-constraint replicas to hard-constraint ones. This approach directly addresses the constraint satisfaction barrier, demonstrating $O(N^5)$ gap improvement in time-to-solution on benchmark problems. The non-reversible DEO swap schedule of Syed et al. (arXiv:2102.07720, JRSS-B 2022) provides the optimal annealing protocol, with round-trip rate converging to $(2+2\Lambda)^{-1}$.

Simulated Quantum Annealing, despite theoretical elegance, **fails on wide barriers**. The Crosson-Harrow result (FOCS 2016, arXiv:1601.03030) proves that SQA's exponential advantage over classical SA applies specifically to *thin, high* barriers via instanton tunneling. [arXiv](#) For barriers spanning 200+ pair flips, the instanton path becomes exponentially long, eliminating any quantum advantage. This definitively rules out SQA as a tunneling strategy for this problem.

Minimal viable experiment: Implement an ejection chain on the constraint graph. From the current local minimum, select the pair with highest LP sensitivity (from dual values). Flip its direction, propagate through 5–50 dependent pairs using BFS, repair via constraint propagation at each depth, evaluate LP at depths 1, 5, 10, 20, 50. Track the best trial solution versus depth. Estimated setup: 1 week.

The constraint graph's eigenvalue structure reveals natural decomposition

The constraint adjacency graph is $L(K_n)$ —the line graph of K_n —a strongly regular graph with parameters $(n(n-1)/2, 2(n-2), n-2, 4)$. Its adjacency spectrum has exactly three distinct eigenvalues: $2n-4$ (multiplicity 1), $n-4$ (multiplicity $n-1$), and -2 (multiplicity $n(n-3)/2$). This known structure is analytically powerful. The $n-1$ eigenvectors at eigenvalue $n-4$ encode “macro-level” coupling—which macro each pair involves—while the overwhelming majority of eigenvalues at -2 represent generic pair interactions. The Laplacian algebraic connectivity is n , and higher-order Cheeger inequalities (Lee, Oveis Gharan, Trevisan, 2014) confirm that the natural k -way partition has $k \leq n$ parts, corresponding exactly to individual macros.

The practical value lies not in the raw $L(K_n)$ spectrum but in the **weighted** constraint graph where weights reflect LP dual values and current assignment sensitivity. Computing the top-20 Laplacian eigenvectors of this weighted graph (routine with ARPACK for 20K–144K vertices) reveals which groups of macro pairs can be flipped quasi-independently. Spectral clustering on this embedding identifies natural “flip neighborhoods” that minimize inter-group constraint coupling.

Practical expander decomposition algorithms, now available from Gottesbüren, Parotsidis, and Probst Gutenberg (ESA 2024, first practical implementation), can identify bottleneck cuts in near-linear time $\tilde{O}(m)$. Applied to the effective constraint graph weighted by constraint tightness, these algorithms partition the problem into subproblems connected by narrow bottlenecks—exactly the structure needed for hierarchical decomposition. The spectral gap of subgraphs induced by proposed flip sets directly predicts feasibility: if the gap drops below a threshold, circular constraint chains become likely.

For matrix sketching and cheap LP bounds, recent work by Çekirge, Gay, and Woodruff (APPROX/RANDOM 2025) provides multipass linear sketches for LP-type problems with space polynomial in dimension and polylogarithmic in $1/\epsilon$. A Clarkson-Woodruff sparse sketch can compress the constraint matrix from $O(n^3)$ rows to $O(n^2 \cdot \text{polylog})$ rows, yielding $(1+\epsilon)$ -approximate LP bounds in time $O(\text{nnz}(A))$ —seconds rather than the minutes of a full LP solve.

Minimal viable experiment: Compute the weighted Laplacian of the constraint graph using LP dual values as edge weights. Extract top-20 eigenvectors. Apply k -means clustering. Test whether resulting clusters align with macro groups that can be optimized independently by solving separate LPs for each cluster. Compare total LP cost (sum of cluster LPs) to full LP. Estimated setup: 2–3 days.

A four-level evaluation cascade screens 99.9% of candidates at negligible cost

The research across Areas A, D, and F converges on a concrete multi-fidelity cascade architecture:

- **Level 0 (nanoseconds):** Zobrist hashing for duplicate detection. Assign 4 random 64-bit numbers per macro pair (one per direction). Hash of full assignment = XOR of all table entries. Incremental update on single-pair flip: one XOR operation. [Wikipedia](#) Maintain a transposition table of all evaluated assignments.
- **Level 1 (microseconds to milliseconds):** Structural feasibility filters. Cycle detection in the induced precedence graph via Bellman-Ford in $O(V+E)$. Edge-finding and not-first/not-last constraint propagation rules from disjunctive scheduling [ScienceDirect](#) (Dorndorf, Pesch & Phan-Huy, AI 2000) [Springer](#) in $O(n^2)$. These immediately reject assignments creating infeasible precedence cycles without any LP computation.
- **Level 2 (microseconds):** Dual-based cluster bounds plus surrogate model screening. Use the Miftari et al. Lagrangian bounds from previously solved LPs. Supplement with a Random Forest surrogate (SMAC3-style) trained on all LP evaluations to date. Features include aggregate assignment statistics, graph-theoretic properties of the precedence graph, and spectral features from the constraint graph Laplacian. Reject candidates whose predicted LP cost exceeds the best known solution.
- **Level 3 (50–150ms):** Partial LP evaluation. Solve the LP with 10–20% of constraints (strategically selected via the screening rules), or run 10–50 simplex iterations with early termination. The constraint propagation-based LP bounding method of Daskin and Werner (JAIR 2024) provides an alternative: applying arc consistency to complementary slackness conditions yields LP bounds “often not far from the global optimum” at linear space complexity. [Journal of Artificial Intelligence Res](#) Only candidates surviving this filter proceed to full LP.
- **Level 4 (seconds):** Full LP solve with warm-starting from the nearest previously solved LP. Multi-fidelity Knowledge Gradient (Wu & Frazier, UAI 2019; implemented in BoTorch) selects which candidate to evaluate at which fidelity level, jointly optimizing the information gain per unit cost. The budget allocation: 5 random initial evaluations to train the surrogate, 5 partial LP evaluations to calibrate fidelity correlation, then 20 full LP evaluations guided by MFKG.

A critical enabling step is identifying “don’t-care” macro pairs—those whose assignment doesn’t affect the LP cost. LP sensitivity analysis via dual values identifies these directly: pairs with zero dual values for all assignment constraints can be fixed arbitrarily, [MIT](#)

[Ucsd](#) reducing the effective search space from 4^m to $4^{m'}$ where $m' \ll m$. This echoes

the “safe screening rules” from sparse optimization (Ndiaye et al., arXiv:1611.05780), adapted from eliminating LASSO features to eliminating combinatorial variables.

Graphs of Convex Sets and Dead-End Elimination: two powerful cross-domain transfers

The closest structural analog is **Graphs of Convex Sets** (GCS) from robotics motion planning (Marcucci et al., SIAM J. Optimization 2024, arXiv:2101.11565; Science Robotics 2023, arXiv:2205.04422). GCS formalizes exactly the “union of convex sets” structure: a graph where each vertex is paired with a convex program, edges couple programs through additional costs/constraints, [MIT DSpace](#) and the shortest path problem jointly optimizes the discrete path choice and continuous variables within each set. [arXiv](#) The key innovation is a **perspective-based convex relaxation** of the resulting MICP [arXiv](#) that is empirically very tight—often exact—enabling solution via a single convex program followed by depth-first rounding.

The scale challenge is severe: GCS has been demonstrated on graphs with up to 2,500 vertices and 70-dimensional variables, while macro placement has $4^{20,000}$ polyhedra. Direct application is impossible. However, the implicit search extensions—GCS* (arXiv:2407.08848) and lxG (arXiv:2410.08909)—interleave graph search with local convex optimization, exploring only a fraction of the graph. Adapting these implicit search methods to macro placement—where adjacency means “differing in one constraint direction” and the LP serves as the convex optimization oracle—is the most promising cross-domain transfer, requiring engineering of the implicit graph representation and efficient local relaxation computation.

Dead-End Elimination from protein side-chain packing (OSPREY software, Donald Lab; Gainza et al. 2012) provides a complementary pruning paradigm. DEE proves that a rotamer r_i at position i can be eliminated if its minimum possible contribution (over all other positions’ rotamers) exceeds another rotamer’s maximum possible contribution. Adapted to macro placement: for pair (i,j) , constraint direction d can be eliminated if choosing d always leads to higher LP cost than direction d' , regardless of all other assignments. Computing these bounds requires pairwise interaction estimates—analogueous to the precomputed residue-residue energies in protein design. The challenge is that protein interactions are local (each residue interacts with ~10–50 neighbors) while macro placement coupling is global through the LP. **Sparsification via spectral clustering** bridges this gap: after decomposing the constraint graph into weakly-coupled clusters, pairwise interaction bounds become computationally tractable within each cluster.

The **complementarity formulation** from multi-body contact mechanics (Yang et al., IEEE

Trans. Robotics 2024; Pang et al., MIT/Tedrake Lab 2023) represents non-overlap constraints as complementarity conditions ($\text{gap} \geq 0$, $\text{force} \geq 0$, $\text{product} = 0$). Interior-point methods for the resulting linear complementarity problem implicitly navigate between discrete contact modes along a central path, providing smooth traversal of the discrete topology space. While the user explicitly rejects continuous relaxation for the final solution, the complementarity LP provides a powerful heuristic for generating candidate discrete assignments: solve the smoothed problem, then “snap” the complementarity variables to discrete modes.

Backdoor variables and AD^3 decomposition for the dense constraint graph

Tree decomposition and survey propagation both fail on the full problem for fundamental reasons: [Springer](#) K_n has treewidth $n-1$, and the cavity method assumes locally tree-like structure [Wikipedia](#) that K_n entirely lacks. However, **backdoor variable theory** (Ganian, Ramanujan, Szeider, STACS 2017) [Springer](#) provides the conceptual bridge. A strong backdoor set B is a subset of pair-assignment variables such that fixing B makes the remaining problem tractable. [AAAI](#) If $|B| = 30\text{--}50$ critical pairs can be identified (via LP dual sensitivity analysis or fractional relaxation), the residual constraint graph after conditioning on B may have bounded treewidth, enabling exact dynamic programming on the tree decomposition.

The **STP decomposability theorem** (Dechter, Meiri, Pearl, AIJ 1991) provides essential theoretical support. For a fixed discrete topology, the continuous placement is exactly a Simple Temporal Problem—a conjunction of difference constraints $x_j - x_i \in [a_{ij}, b_{ij}]$, solvable in $O(n^3)$ by shortest paths. [Springer](#) The decomposability theorem guarantees that any partial assignment satisfying shortest-path constraints extends to a full solution. [Ucla](#) This means topology can be built incrementally: fix pair assignments one at a time, checking feasibility via incremental shortest-paths at each step.

For the full optimization problem, **AD^3** (Alternating Directions Dual Decomposition; Martins et al., JMLR 2015) offers the most scalable MAP inference framework. AD^3 decomposes the factor graph into overlapping tractable subproblems (trees or stars) solved in parallel, with ADMM consensus ensuring global consistency at $O(1/\epsilon)$ convergence. Bethe-ADMM (Fu et al., arXiv:1309.6829) extends this to 7+ million variables using MPI parallelism. For $n=537$ macros with $\sim 144,000$ pairwise factor variables of domain size 4, AD^3 is well within computational reach. The main challenge is specifying factor potentials: these must approximate LP objective contributions for each pair assignment, which could be estimated from the dual-based bounds or surrogate models described above.

The recent FOCS 2024 breakthrough on efficient fractional hypertree width approximation (Korchemna et al., arXiv:2409.20172) makes practical hypertree decomposition feasible for the first time. [arXiv](#) [IEEE Xplore](#) While the full constraint hypergraph has prohibitive width, a sparsified backbone—identified via spectral decomposition or LP sensitivity—may have manageable width, enabling exact optimization on the backbone with heuristic extension to remaining pairs.

Conclusion: a six-week implementation roadmap

Three insights cut across all six research areas. First, **the evaluation bottleneck is solvable**: the Miftari dual bounds, constraint propagation filters, and partial LP fidelity levels can screen candidates 100–10,000× faster than full LP solves, expanding the effective evaluation budget from ~30 to thousands of candidates. Second, **the infeasibility barrier is an engineering problem, not a theoretical one**: ejection chains and ALNS destroy-and-repair traverse infeasible intermediates by design, [ScienceDirect](#) transforming the “wide barrier” from an obstacle into a structured search space. Third, **the constraint graph’s spectral structure provides the natural decomposition for free**: the $L(K_n)$ eigenvalue spectrum analytically identifies macro-level groupings, while weighted spectral clustering reveals which flip neighborhoods are quasi-independent.

The most speculative ideas with strong mathematical connections are the GCS perspective relaxation adapted to implicit graph search (from robotics, ~2 years from maturity for this scale) and DEE-style pruning of constraint directions (from protein design, requiring novel pairwise interaction bound computation). The most immediately implementable are the Zobrist hashing + constraint propagation cascade (days), LP warm-starting with dual bounds (3–5 days), and ejection chains with spatially-coherent neighborhoods (1 week).

For a 6-week competition timeline, the recommended sequence is: **Week 1**: Implement the evaluation cascade (Levels 0–2) and Zobrist hashing. **Week 2**: Build ejection chain infrastructure with constraint-propagation repair. **Week 3**: Add LP warm-starting and Miftari dual bounds for cluster pruning. **Week 4**: Compute spectral decomposition to guide neighborhood selection; integrate ALNS with adaptive operator selection. **Week 5**: Train Random Forest surrogate on accumulated LP evaluations; implement multi-fidelity Knowledge Gradient for evaluation allocation. **Week 6**: Integrate, tune, and benchmark against competition baselines. This roadmap prioritizes methods that are mature (HIGH maturity rating), scale to $n=200\text{--}537$ (demonstrated on comparable problem sizes), and directly address the stated constraints (no continuous relaxation, legal states by construction, ~30 LP evaluation budget).